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Superconducting Radio-frequency Cavities Made from Medium and Low-Purity Niobium Ingots

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Abstract.

Superconducting radio-frequency cavities made of ingot niobium with residual resistivity ratio (RRR) greater than 250 have proven to have similar or better performance than fine-grain Nb cavities of the same purity, after standard processing. The high purity requirement contributes to the high cost of the material. As superconducting accelerators operating in continuous-wave typically require cavities to operate at moderate accelerating gradients, using lower purity material could be advantageous not only to reduce cost but also to achieve higher Q_0 -values. In this contribution we present the results from cryogenic RF tests of 1.3-1.5 GHz single-cell cavities made of ingot Nb of medium (RRR=100-150) and low (RRR=60) purity from different suppliers. Cavities made of medium-purity ingots routinely achieved peak surface magnetic field values greater than 70 mT with an average Q_0 -value of 2×10^{10} at 2 K after standard processing treatments. The performances of cavities made of low-purity ingots were affected by significant pitting of the surface after chemical etching.

1. Introduction

Superconducting radio-frequency (SRF) cavities are used in modern particle accelerators to efficiently accelerate a charged particle beam. The high efficiency is due to the power dissipated in the cavity wall being several orders of magnitude lower for a superconductor than for a normal conductor. The vast majority of SRF cavities are made of niobium, either bulk or as a thin film deposited on copper.

The performance of SRF cavities is described by a plot of the quality factor, Q_0 , as a function of the accelerating gradient, E_{acc} , or as a function of the peak surface magnetic or electric fields, B_p and E_p , respectively, measured at or below 4.2 K. Research and development efforts throughout the world over the last decade demonstrated that the performance of SRF cavities built from large-grain, ingot material is comparable to that achieved using standard fine-grain (ASTM ≥ 5), bulk Nb [1]. The average grain diameter for ingot material is greater than about 1 cm, compared to $\sim 50 \mu\text{m}$ for standard fine-grain Nb. A relatively small production series of 1.3 GHz, 9-cell cavities made of ingot Nb material showed that high gradients, greater than $\sim 40 \text{ MV/m}$, could be achieved reproducibly and that higher Q_0 -values than those of cavities of the same shape made of standard fine-grain Nb, were achieved at 2 K [2].

The residual resistivity ratio (RRR) is a parameter used to specify the purity of the metal and is proportional to the thermal conductivity at 4.2 K. We refer to high-purity Nb when the RRR is greater than ~ 250 , medium-purity for $100 \leq \text{RRR} \leq 250$ and low-purity (or reactor grade) for $\text{RRR} < 100$. The purity of Nb is mostly determined by the concentration of interstitial impurities such as H, C, N and O and it is increased by increasing the number of melting cycles of the Nb ingots inside vacuum electron-beam melting furnaces [3]. The purity of the Nb commonly specified for use in SRF cavity fabrication is $\text{RRR} \geq 300$.

The push for higher purity material resulted from the need to increase the thermal conductivity of the Nb in order to “thermally stabilize” defects in the material, which can cause quenches at reduced gradients [4]. The production of high-purity, fine-grain Nb sheets requires many elaborate steps during which foreign metallic impurities can be embedded in the material and therefore also requiring time-consuming quality control of the sheets [5]. The cost of high-purity, fine-grain Nb sheets is significant and exceeds ~600 \$/kg. The production of ingot Nb discs is comparatively much simpler, as it only requires slicing of the Nb ingot, and therefore lower cost is to be expected. Whereas SRF cavities for high-energy particle colliders need to operate at gradients corresponding to B_p -values greater than ~100 mT, SRF cavities for continuous wave (CW) accelerators require operation at 70-80 mT but with high quality factor ($> \sim 1 \times 10^{10}$ at 2 K) [6].

The purity of the Nb material is a parameter which affects cost, quench field, surface resistance and the mechanical properties of the cavity. The mechanical properties of medium-purity ingot Nb cavities have been recently evaluated and the yield pressure was found to be higher than that of high-purity, fine-grain Nb cavities, for the same annealing temperature [7].

In this contribution, we present the results of systematic studies of 1.3-1.5 GHz single-cell elliptical cavities made of ingot Nb material with RRR in the range 60-150 to evaluate whether an “optimum” RRR from the cost-performance point of view can be found. The treatment processes included standard buffered chemical polishing (BCP), electropolishing (EP), centrifugal barrel polishing (CBP), high-temperature annealing in vacuum (HT), low-temperature baking (LTB), HF rinse (HF), and the recently developed titanium [8] and nitrogen [9] doping.

2. Niobium material and cavity fabrication

Three medium-purity Nb ingots were provided by CBMM, Brazil. Discs of thickness 3.175 mm were sliced from the ingots by wire electrodischarge machining. Four low-purity discs, 2.8 mm thick, were purchased from Ningxia, China. The low purity discs were made by saw-cutting an ingot, followed by thickness reduction on a lathe and mechanical polishing. The discs were annealed at 800°C for 2 h in a vacuum furnace to reduce the hardness. The low-purity discs were also scanned for defects applying the eddy-current method at DESY and no defects were found. Two 2.8 mm thick medium purity ingot Nb discs were purchased from Tokyo-Denkai (TD), Japan. The discs were wire-cut from an ingot. Table 1 shows the material supplier, the concentration of the main interstitial impurities, the RRR and the Vickers hardness. The RRR of the medium-purity ingots was calculated from the resistance of $1 \times 1 \times 30 \text{ mm}^3$ samples measured with the standard four-probe method with an estimated error of 10%. The RRR calculated from the thermal conductivity at 4 K measured on another set of samples from ingots F, G, H was 126, 144 and 93, respectively, whereas the thermal conductivity measurement at 4 K of the low-purity samples resulted in a RRR-value of 60, the same as that given by the manufacturer.

The cavity fabrication followed the standard practice of deep-drawing of the discs, followed by trimming, machining of the iris and equator of the half-cells, electron-beam welding (EBW) of the cut-off tubes (made of low-purity fine-grain Nb) to the irises and final EBW at the equator. The regions to be welded are chemically etched to remove ~20 μm of material after machining. A list of the cavities built from the different material is listed in table 2 along with the treatment of the discs prior to deep drawing. The 1.3 GHz cavities are of the TESLA center-cell shape [10], whereas the 1.5 GHz cavities are of the original CEBAF shape [11]. A picture of the cavities built and tested for this study is shown in figure 1.

Table 1. Contents ppm (per weight) of the main interstitial impurities from the different Nb ingots and RRR obtained from the samples’ thermal conductivity at 4.2 K.

Ingot label	Supplier	Ta	H	C	O	N	RRR	Vickers hardness
F	CBMM	~1350	~0.3	~1	~9	~8	118 ± 11	54 ± 2
G	CBMM	~1350	~0.3	~1	~9	~8	125 ± 20	55 ± 3
H	CBMM	~700	~0.3	~1	~9	~8	100 ± 10	58 ± 3
n/a	Tokyo-Denkai	29	<5	<10	<10	<10	140 ± 10	64 ± 2
n/a	Ningxia	<100	3	9	110	30	60 ± 6	60 ± 2

Table 2. Cavities’ label and resonant frequency after fabrication, Nb supplier and processing done to the discs prior to deep-drawing or to the half-cells after deep-drawing.

Cavity label	Supplier/Ingot	Discs/Half-cell processing	Frequency (GHz)
F1F2	CBMM/F	none	1.293
F3F4	CBMM/F	Discs: 10 µm BCP, 800 °C/3 h HT	1.466
F5F6	CBMM/F	Discs: 10 µm BCP, 800 °C/3 h HT	1.464
G1G2	CBMM/G	None	1.472
		Discs: 5 µm BCP	
H1H2	CBMM/H	Half-cell H1: 70 µm BCP	1.466
		Half-cell H2: 800 °C/5 h HT	
H3H4	CBMM/H	Half-cells: 20 µm BCP, 1200 °C/3 h HT	1.467
LG-RG	Tokyo-Denkai	None	1.468
TC1N1	Ningxia	Discs: 20 µm BCP, 800 °C/3 h HT	1.287
OCC1N2	Ningxia	Discs: 20 µm BCP, 800 °C/3 h HT	1.463



Figure 1. Picture of the medium and low-purity ingot Nb single-cell cavities used for the study.

3. Experimental methods

The treatment processes applied to the cavities before testing include BCP, EP, CBP, HT, LTB, HF and nitrogen or titanium doping. BCP is done using a mixture of electronic-grade HF(49%):HNO₃(70%):H₃PO₄(85%) = 1:1:1 or 1:1:2 by volume, at a temperature of 15-20 °C. EP is done in a horizontal, rotating setup, using a mixture of electronic grade HF(49%):H₂SO₄(96%) = 1:9 at a

constant voltage of ~ 14 V, a temperature of 15 - 20 °C and a speed of 1 rpm. The acid flow is ~ 3.3 l/min. HF rinse is done by filling the cavity with HF(49%) at 20 °C for 5 min. After BCP, EP or HF, the cavities are thoroughly rinsed with ultra-pure water. CBP is done in three steps using a “coarse” media, (AH-41 ceramic mix, Mass Finishing Inc., ~ 8 $\mu\text{m/h}$ removal rate) an intermediate media (RG-22 cones, Mass Finishing Inc., ~ 2 $\mu\text{m/h}$ removal rate) and a final step with treated corn cobs (MF-3, Mass Finishing Inc.) to achieve a mirror-finish surface. For the first two steps, the cavity is filled with the polishing media, water and a detergent, whereas it is dry for the last step. The cavity rotates inside the CBP machine at 115 rpm. After the first two polishing steps, the cavity is pressure-washed and degreased with water and a detergent in an ultrasonic tank. HT is done in a high-vacuum furnace (either resistive or induction type). LTB is done by flowing hot air around the cavity inside a portable oven, while the inner volume of the cavity is kept below $\sim 10^{-7}$ mbar, to heat the cavity to 120 °C for several hours. Nitrogen doping is done by annealing the cavity at 800 °C/ 3 h in the high-vacuum furnace, injecting high-purity N_2 gas to maintain a pressure of ~ 20 - 25 mTorr for few minutes, pumping-out the N_2 gas and continue annealing for few minutes before turning-off the furnace heaters. Few micrometers of Nb are removed from the inner surface of Nb by EP after N-doping as part of the process, resulting in a nitrogen concentration of ~ 100 at. ppm within few microns from the surface. Doping with Ti was done by annealing the cavity in a high-vacuum induction furnace above 1200 °C for a few hours, resulting in a Ti concentration of ~ 1 at.% within the top ~ 100 nm. The source of Ti was from the Ti45Nb cavity flanges.

Common procedures applied to all cavities include high-pressure (~ 8 MPa) rinse with ultra-pure water, drying in an ISO4/5 clean room, assembly of flanges with RF feedthroughs and antennae for cavity excitation and field probe, as well as a pump-out port. The cavities are evacuated to $\sim 10^{-8}$ mbar before being inserted in a vertical cryostat filled with liquid He at 4.3 K. The residual DC magnetic field inside the cryostat is ~ 5 mG. A plot of the temperature at the top and bottom iris of a cavity during a typical cooldown from 300 K is shown in figure 2. The cooling rate when the top and bottom cavity temperature reaches ~ 9 K is ~ 3 K/min. The temperature difference between the top and bottom iris (~ 10 cm apart), when the bottom iris temperature reaches ~ 9 K, is ~ 4 K.

The cavity immersed in the He bath at 4.3 K is excited using a phase-locked loop. A measurement of the incident, reflected and transmitted power along with the resonant frequency and the decay time allow calculating Q_0 and B_p [12]. The cavity quality factor is measured at a constant, low RF field ($B_p \sim 5$ mT) as the temperature is lowered to 1.6 - 1.5 K by pumping on the He bath. The He bath temperature, T , is measured with a calibrated germanium resistance-temperature device. The average surface resistance, R_s , is calculated as $R_s = G/Q_0$, where G is a geometry factor which is calculated from each cavity shape ($G = 273 \Omega$ for the CEBAF shape, $G = 270 \Omega$ for the TESLA shape). The $R_s(T)$ data are fitted with the following equation:

$$R_s(T) = R_{BCS}(T, \Delta/kT_c, \ell) + R_{res} \quad (1)$$

where R_{res} is the so-called residual resistance and R_{BCS} is the so called Bardeen-Cooper-Schrieffer (BCS) surface resistance. Δ is the energy gap at 0 K, k is the Boltzmann’s constant, T_c is the critical temperature, ℓ is the normal-electrons’ mean free path. R_{BCS} is obtained from a numerical calculation of the surface impedance derived from the BCS theory of superconductivity, valid at low RF field [13]. The fit parameters are R_{res} , Δ/kT_c and ℓ . $T_c = 9.25$ K, $\lambda_L = 32$ nm (London penetration depth) and $\xi_0 = 39$ nm (coherence length) are considered material constants of Nb.

Measurements of $Q_0(B_p)$ are done at 2.0 K up to the breakdown field, $B_{p,max}$. The typical experimental uncertainties are 10% and 5% for Q_0 and B_p measurement, respectively. For some of the cavity tests, temperature mapping [14, 15] or oscillating superleak transducers [16, 17] were used to determine the breakdown location. The breakdown location was inspected after the cavity is warmed-up to room temperature using a high-resolution inspection camera inserted into the cavity [18].

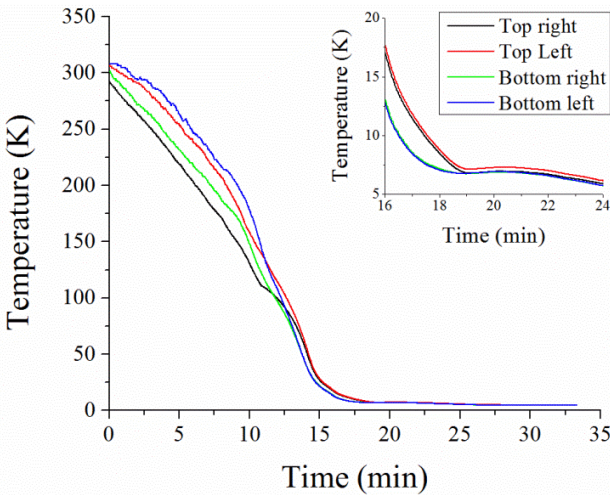


Figure 2. Temperature measured at the top and bottom iris of a cavity at two locations 180° apart during a typical cool-down from 300 K to 4.3 K.

4. Cavity test results

4.1 Medium-purity Nb cavities

The treatment processes prior to each test are listed in table 3 for each of the cavities. HT above 1200°C refers to Ti doping, while “2N10+7μm EP” refers to N-doping. “2N10” means that the N₂ was injected for 2 min, followed by annealing in vacuum at 800 °C for 10 min. Figure 3 shows the evolution of the quench field and of the quality factor at 2.0 K at low field (18 mT) for different treatments after fabrication for cavity LG-RG. No field emission was detected in any of the tests. The reduction of Q_0 as the field increased from 18 mT to 70 mT is <10% on average.

Table 3. Treatment processes applied to medium-purity ingot Nb cavities prior to each test.						
Test No.	LG-RG	F1F2	F3F4	G1G2	H1H2	H3H4
1	100μm BCP, 800°C/2h, 50μm BCP	105μm CBP, 50μm BCP, 800°C/2h, 20μm BCP	100μm CBP, 50μm BCP, 800°C/2h, 20μm BCP	70μm CBP, 65μm BCP, 800°C/3h, 20μm BCP	70μm BCP, 600°C/10h	130μm CBP, HF rinse, 800°C/3h
2	120°C/48h LTB	120°C/12h LTB	120°C/12h	800°C/6h	1400°C/3h	30μm EP
3	1250°C/3h, 10μm BCP	25μm EP	HF rinse	120°C/24h	100μm CBP, 800°C/2h, 48μm BCP	10μm BCP

4	50µm BCP	120°C/12h LTB	30µm EP	15µm BCP	20µm EP	20µm EP
5	re-HPR	re-HPR	120°C/12h	1000°C/6h	20µm BCP	60µm EP
6	1400°C/3h	50µm BCP	HF rinse	120°C/12h	--	--
7	8µm BCP	25µm BCP	--	29µm BCP	--	--
8	120°C/15h LTB	6µm BCP	--	1200°C/6h	--	--
9	HF rinse	120°C/48h LTB	--	120°C/12h	--	--
10	2N10, 7µm EP	HF rinse	--	20µm BCP	--	--
11	2µm BCP	25µm EP	--	1400°C/3h	--	--
12	40µm CBP, 20µm BCP, 800°C/3h, 30µm BCP	120°C/24h LTB	--	120°C/12h	--	--
13	26µm BCP	HF rinse	--	HF rinse	--	--
14	25µm BCP	1N5, 5µm EP	--	10xHF rinse, 100nm oxypolishing 30µm EP	--	--
15	50µm BCP	--	--	1400°C/3h	--	--
16	120°C/12h LTB	--	--	1µm BCP	--	--
17	HF rinse	--	--	1400°C/3h	--	--
18	20µm EP	--	--	25µm BCP	--	--
19	120°C/12h LTB	--	--	1250°C/3h	--	--
20	HF rinse	--	--	10µm BCP, 1300°C/3h	--	--
21	1N5, 5µm EP	--	--	15µm BCP, 2N10, 7µm EP	--	--

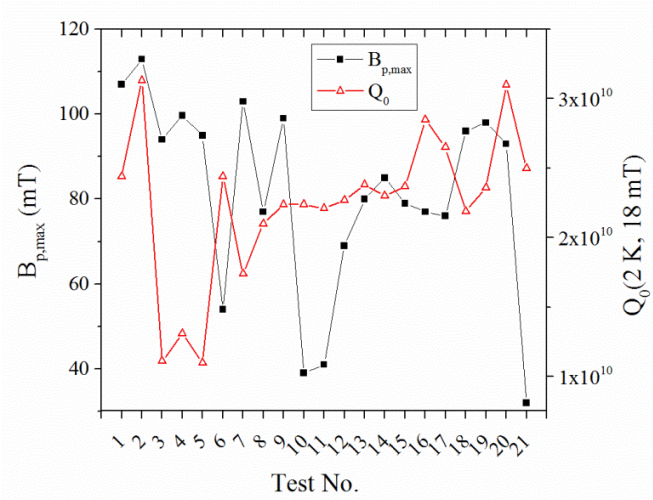


Figure 3. $B_{p,max}$ and $Q_0(2\text{ K}, 18\text{ mT})$ measured for cavity LG-RG for different treatments, listed in table 3.

Figure 4 (a) shows the evolution of the maximum field and $Q_0(2\text{ K})$ at low field for different treatments applied to cavity F1F2 after fabrication. Field emission did not occur during any of the tests, however, strong multipacting (MP) inducing quenches often occurred in the range 70-100 mT. Tests where the maximum field was limited by MP are indicated by arrows in figure 4 (a). A similar plot to that in figure 4 (a) is shown in figure 4 (b) for cavity F3F4, which was also limited by quench. No field emission was present in any of the tests, although MP was encountered at $\sim 110\text{ mT}$ during the tests No. 4 and No. 6. The quench field of cavity F3F4 was at the equator region but the azimuthal location was different after BCP and after EP. The cavity F5F6 was tested once after $100\text{ }\mu\text{m}$ removal by BCP, $50\text{ }\mu\text{m}$ removal by BCP, HT at $800\text{ }^\circ\text{C}/2\text{ h}$, and $20\text{ }\mu\text{m}$ removal by BCP. The quench field of F5F6 was $(85 \pm 6)\text{ mT}$ and the $Q_0(2\text{ K}, 70\text{ mT})$ was $(1.7 \pm 0.3) \times 10^{10}$.

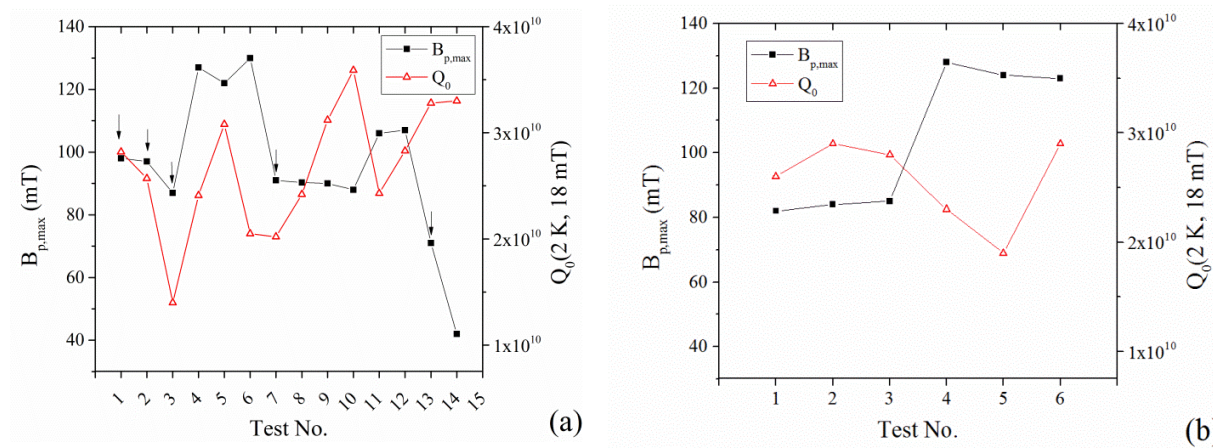


Figure 4. $B_{p,max}$ and $Q_0(2\text{ K}, 18\text{ mT})$ measured for cavity F1F2 (a) and F3F4 (b) after different treatments, listed in table 3. Arrows indicate tests with MP-induced quenches.

Figure 5 shows the evolution of the maximum field and $Q_0(2\text{ K}, 18\text{ mT})$ for different treatments applied to cavity G1G2 after fabrication. There was no field emission in any of the tests. However, MP at $\sim 80\text{ mT}$ was sometimes observed. It could have induced quenches in the tests marked by arrows in figure 5. The treatments “ $1250\text{ }^\circ\text{C}/3\text{ h}$ ”, “ $1300\text{ }^\circ\text{C}/3\text{ h}$ ” and “ $1400\text{ }^\circ\text{C}/3\text{ h}$ ” resulted in doping of the Nb surface

with Ti [19]. The quench location, determined by temperature maps taken during the power rise at 2 K, changed after BCP/EP but did not change after HT. In all cases, no significant heating was measured at the quench location at the maximum field, just below quench, as the local temperature rise was less than 6 mK in all tests. Except for the quench during test 11 which was found to be close to a grain boundary, no outstanding features were noticed by optical inspection at the other quench locations.

Figure 6 shows the evolution of the maximum field and $Q_0(2\text{ K}, 18\text{ mT})$ for different treatments applied to cavities H1H2 and H3H4 after fabrication. There was no field emission in any of the tests, which were limited by quench. Temperature mapping during test 1 showed the quench to be in half-cell H1, in a region where “etch pits” were visible. The following annealing at $1400\text{ °C}/3\text{ h}$ improved the Q_0 -value but the cavity quenched at lower field at the same location. A different quench location was found during test 3 with no noticeable feature visible by optical inspection following the RF test. Temperature maps taken during high-power tests of H3H4 showed that the quench location changed after additional chemistry. Optical inspection showed isolated pits visible at the quench spots, except after the last test, after $60\text{ }\mu\text{m}$ EP, where a cluster of pits was present. Figure 7 (a) shows a map of the cavity surface with the position of the temperature sensors and the quench locations for both H1H2 and H3H4. Figure 7 (b) shows the temperature rise over the He bath temperature at 2 K as a function of the local surface magnetic field at the quench locations: a small overheating can be seen in most of the cases. An image of the cluster of pits found at the quench location at the equator of H3H4 after test 5, obtained with a high-resolution optical inspection system, is shown in figure 8 (a). A picture of the half-cell H1, during fabrication of cavity H1H2, is shown in figure 8 (b). Etch-pits became visible after $\sim 70\text{ }\mu\text{m}$ material removal by BCP and the quench location during tests 1 and 2, after the cavity fabrication was completed, is encircled in red in figure 8 (b). Measurements of the thickness along the profile of the half-cells showed that half-cell H1 was significantly thicker than H2 and also had a thick spot at a grain boundary [20]. Such thick spot could have “grabbed” the die during forming, introducing large quantities of dislocations, which were revealed by the formation of pits after subsequent chemical etching. The etch pits’ pattern shown in figure 8 (b) is consistent with this explanation.

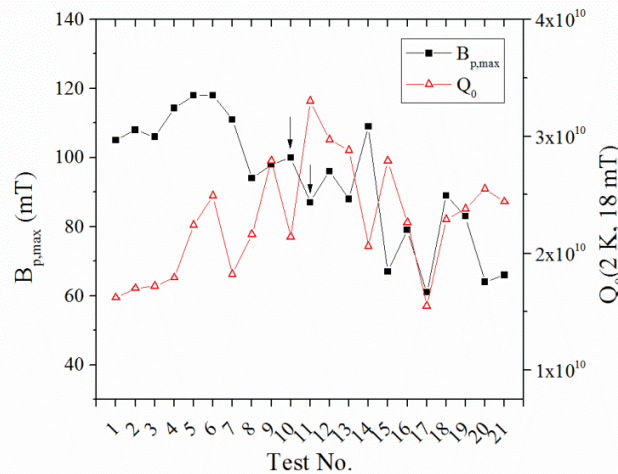


Figure 5. $B_{p,max}$ and $Q_0(2\text{ K}, 18\text{ mT})$ measured for cavity G1G2 after different treatments, listed in table 3. Arrows indicate tests with MP-induced quenches.

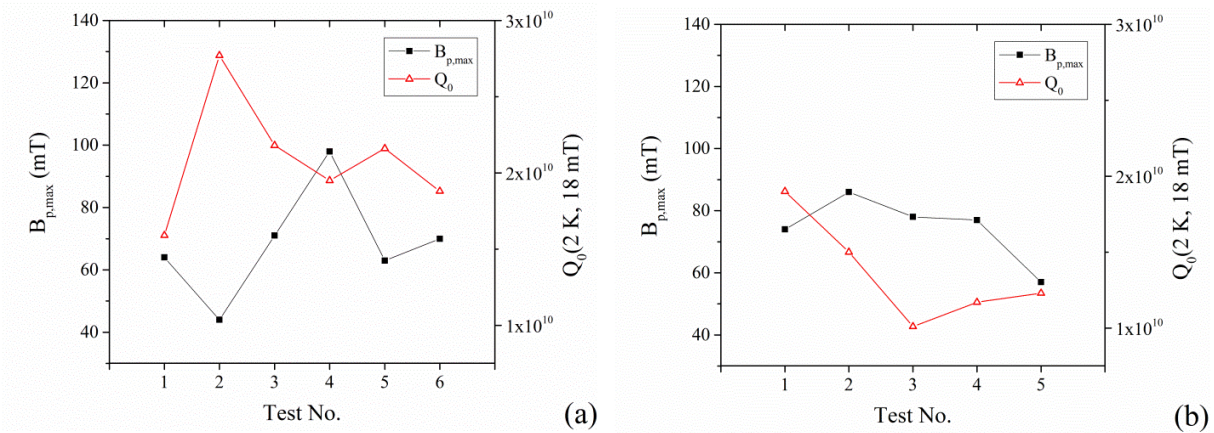


Figure 6. $B_{p,max}$ and $Q_0(2\text{ K}, 18\text{ mT})$ measured for cavity H1H2 (a) and H3H4 (b) after different treatments, listed in table 3.

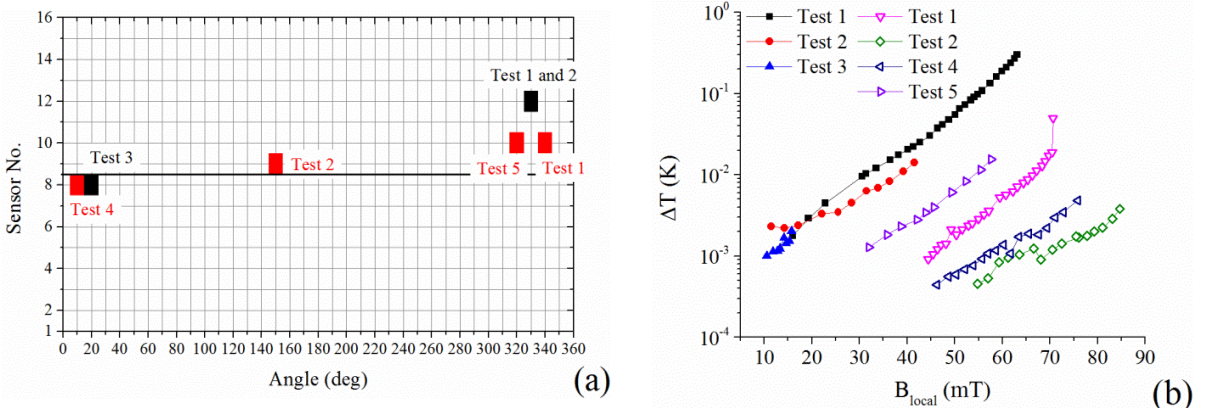


Figure 7. (a) Quench locations found in cavities H1H2 (black squares) and H3H4 (red squares) for different tests, after the treatments listed in table 3. Sensor 1 is at the top iris and sensor 16 is at the bottom iris. The equator weld is between sensors 8 and 9. (b) Temperature rise above the He bath temperature at the quench locations as a function of the local surface field for H1H2 (solid symbols) and H3H4 (empty symbols).

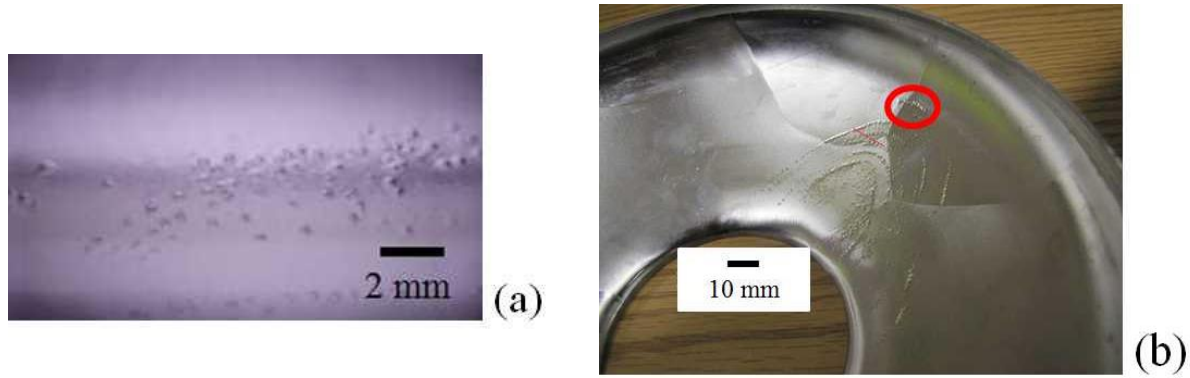


Figure 8. (a) Cluster of etch-pits found at the quench location by optical inspection of the inner surface of cavity H3H4 after test 5. (b) Etch-pits found on half-cell H1 after deep-drawing and 70 μm etching. The quench location of cavity H1H2 after tests 1 and 2 is shown in the red circle.

A plot showing the typical $Q_0(B_p)$ curves measured at 2 K for medium-purity ingot Nb cavities after BCP and EP treatments is shown in figure 9, showing similar Q_0 -values but higher maximum field for cavities treated by EP than BCP. Similar plots after LTB and HF rinse are shown in figure 10. Higher Q_0 -values at low field are obtained by HF rinse, however the Q_0 -values are well within error bars at medium-field. $Q_0(B_p)$ curves after doping with Ti and N (the latter followed by few micrometer removal by EP) are shown in figures 11 (a) and 11 (b), respectively. The increase of Q_0 with increasing field, characteristic of these processes can be seen in figure 10, however the quench field is fairly low (less than 70 mT) in most of the cases.

The weighted average values of the fit parameters R_{res} , Δ/kT_c , ℓ , as well as the weighted average values of the R_{BCS} (4.3 K, 5 mT), Q_0 (2 K, 70 mT) and $B_{p,\text{max}}$ for the different treatments are listed in table 4 for each cavity.

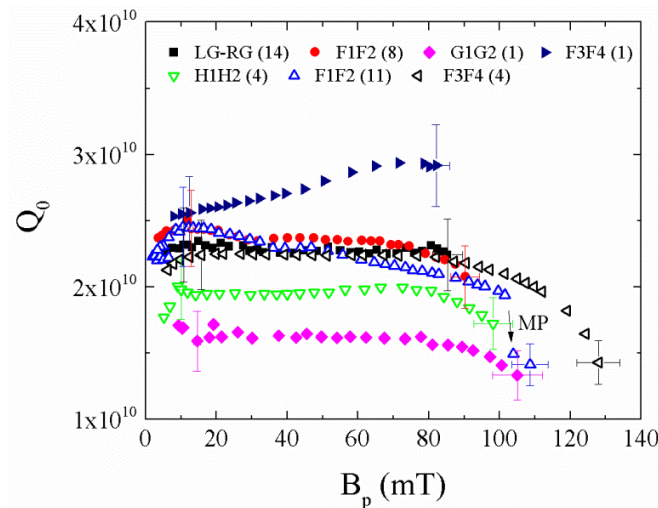


Figure 9. $Q_0(B_p)$ curves at 2.0 K representative of the performance of medium-purity ingot Nb cavities after BCP (solid symbols) and EP (empty symbols) treatments. The numbers between parentheses in the legend refer to the test number. The tests were limited by quench, with no field emission.

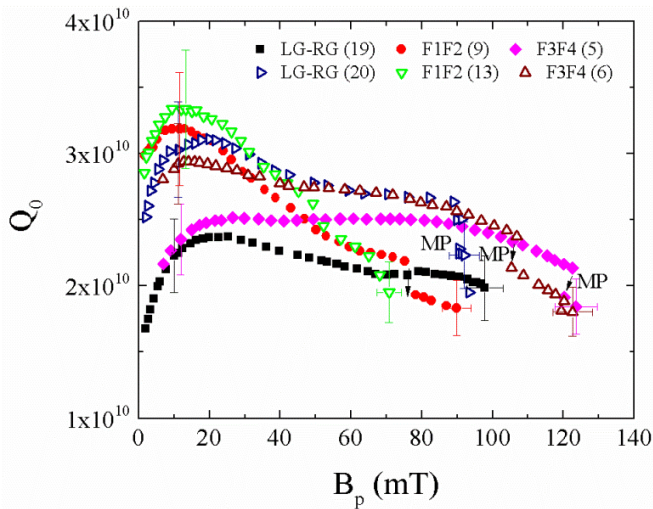


Figure 10. $Q_0(B_p)$ curves at 2.0 K representative of the performance of medium-purity ingot Nb cavities after LTB (solid symbols) and HF rinse (empty symbols) treatments. The numbers between parentheses in the legend refer to the test number. The tests were limited by quench, with no field emission.

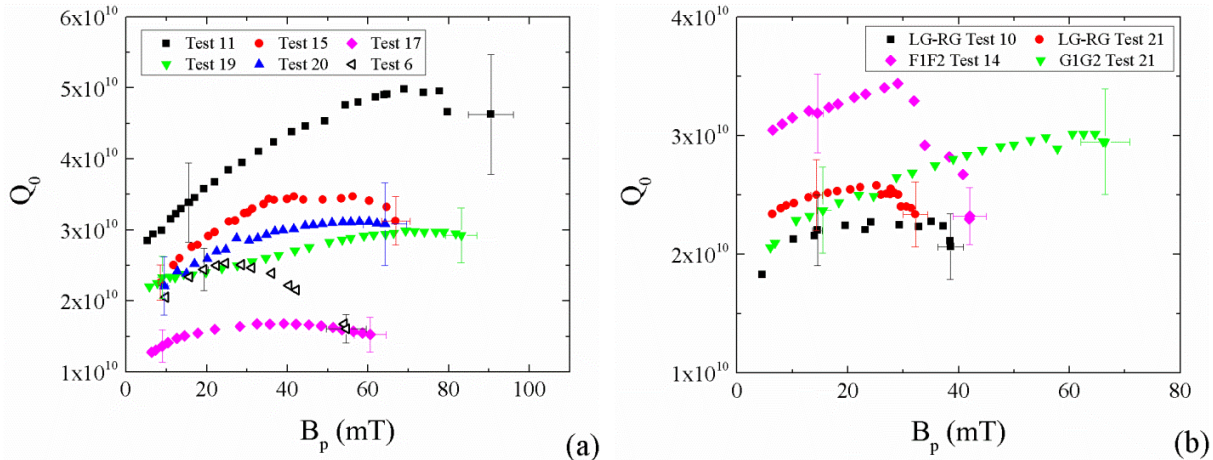


Figure 11. $Q_0(B_p)$ curves measured at 2.0 K after Ti doping of cavity G1G2 (solid symbols) and LG-RG (empty symbols) (a) and N-doping of three different cavities (b). The tests were limited by quench without field emission. The N-doping process on F1F2 and LG-RG, before tests 14 and 21, respectively, was done at the same time.

Table 4. Weighted average values of the fit parameters of $R_s(T)$ and of $R_{BCS}(5 \text{ mT})$ at 4.3 K, $Q_0(70 \text{ mT})$ at 2.0 K and of the maximum peak surface magnetic field.

Cavity	Δ/kT_c	ℓ (nm)	R_{res} (n Ω)	R_{BCS} (n Ω)	$Q_0 (\times 10^{10})$	$B_{p,max}$ (mT)
BCP						
LG-RG	1.81 ± 0.01	86 ± 40	2.1 ± 0.2	697 ± 12	2.1 ± 0.1	84 ± 2
F1F2	1.83 ± 0.01	140 ± 18	2.4 ± 0.1	656 ± 9	2.1 ± 0.1	98 ± 3
F3F4, F5F6	1.837 ± 0.002	27 ± 25	0.4 ± 0.1	656 ± 24	2.2 ± 0.2	83 ± 3
G1G2	1.82 ± 0.01	100 ± 21	1.9 ± 0.1	786 ± 12	2.0 ± 0.1	96 ± 2
H1H2, H3H4	1.81 ± 0.01	95 ± 20	1.6 ± 0.2	776 ± 14	0.9 ± 0.1	70 ± 2
EP						
LG-RG	1.78 ± 0.02	24 ± 110	1.9 ± 0.3	703 ± 20	2.1 ± 0.2	96 ± 4
F1F2	1.84 ± 0.01	138 ± 12	2.5 ± 0.1	619 ± 28	1.6 ± 0.2	100 ± 4
F3F4	1.828 ± 0.002	26 ± 53	1.7 ± 0.1	637 ± 30	2.2 ± 0.3	128 ± 6
G1G2	1.80 ± 0.02	81 ± 26	3.5 ± 0.3	794 ± 26	2.1 ± 0.2	109 ± 5
H1H2, H3H4	1.78 ± 0.01	78 ± 21	8.0 ± 0.2	786 ± 14	1.2 ± 0.1	81 ± 2
LTB						
LG-RG	1.910 ± 0.002	26 ± 44	4.8 ± 0.1	578 ± 11	2.3 ± 0.2	89 ± 3
F1F2	1.89 ± 0.01	36 ± 16	3.8 ± 0.1	446 ± 9	2.1 ± 0.2	99 ± 3
F3F4	1.918 ± 0.004	26 ± 83	3.5 ± 0.1	549 ± 17	2.1 ± 0.2	97 ± 3
G1G2	1.88 ± 0.01	7 ± 1	2.5 ± 0.1	637 ± 13	2.3 ± 0.2	103 ± 4
HF						
LG-RG	1.910 ± 0.003	26 ± 70	2.6 ± 0.1	593 ± 14	2.3 ± 0.2	84 ± 3
F1F2	1.89 ± 0.01	27 ± 50	2.1 ± 0.1	429 ± 9	2.2 ± 0.2	80 ± 2
F3F4	1.895 ± 0.007	27 ± 64	1.9 ± 0.1	540 ± 12	2.9 ± 0.2	97 ± 3
G1G2	1.86 ± 0.01	29 ± 47	1.4 ± 0.2	576 ± 40	4.1 ± 0.6	88 ± 7
HT						
G1G2	1.84 ± 0.01	238 ± 44	2.1 ± 0.2	1020 ± 20	2.1 ± 0.2	104 ± 4
N and Ti Doping						
LG-RG	1.85 ± 0.01	29 ± 26	3.2 ± 0.2	605 ± 12	n/a	35 ± 1
F1F2	1.864 ± 0.002	26 ± 41	2.1 ± 0.1	450 ± 10	n/a	42 ± 2
G1G2	1.86 ± 0.01	79 ± 12	3.7 ± 0.1	673 ± 9	3.2 ± 0.4	71 ± 2

4.1 Low-purity Nb cavities

The sequence of treatments applied to the two low-purity single-cell cavities is listed in table 5. The $Q_0(B_p)$ curves measured at 2.0 K during each test are shown in figure 12. There was no field emission or MP in any of the tests. Severe surface pitting was noticed for both cavities as a result of the BCP etching in preparation for the first test, which showed a very strong degradation of the quality factor with increasing RF field. TC1N1 was etched by BCP 1:1:2, whereas OCC1N2 was etched by BCP 1:1:1, but both showed similar pitting and RF performance. An image of the inner surface of TC1N1 after BCP is shown in figure 13 (a). TC1N1 was re-tested after LTB at 120 °C for 24 h, but there was no effect on the

strong Q-drop. CBP followed by EP were done in order to remove the etch pits and obtain a smooth surface finish. The second test, following these treatments, showed much improved RF performance, although the quench field was still relatively low. Additional EP was done to attempt increasing the quench field without much success. A high-temperature annealing at 1200 °C followed by EP was done to try diffusing any potential surface contamination deeper into the bulk, but no significant improvement was achieved. The values of the fit parameters of $R_s(T)$, of the low-field R_{BCS} (4.3 K, 5 mT) and of the quench field are listed in table 6 for each cavity test. R_{res} is increasing with additional EP. Pitting still occurred after multiple EP, although it did not seem as dense as after BCP. Figure 13 (b) shows an optical image of the inner surface of TC1N1 after the last test. Temperature mapping was applied to OCC1N2 during each RF test and showed the quench occurring always at the same location, about 1.8 cm from the equator. Some pre-heating (ΔT up to ~15 mK) was measured at this location just before quench, in tests 2 and 4. Few pits were visible by optical inspection of the quench location; nevertheless they did not seem as pronounced as other features found on the inner surface, such as pits at grain boundaries or large steps at grain boundaries.

Table 5. Treatment processes applied to low-purity ingot Nb cavities prior to each test.

Test No.	TC1N1	OCC1N2
1	80μm BCP, 800°C/3h, 15μm BCP	60μm BCP, 800°C/3h, 50μm BCP
2	120°C/24h, 70μm CBP, 30μm EP, 800°C/3h, 15μm EP	60μm CBP, 30μm EP, 800°C/3h, 30μm EP
3	30μm EP	30μm EP
4	30μm EP	1200°C/3h, 30μm EP
5	1200°C/3h, 30μm EP	--

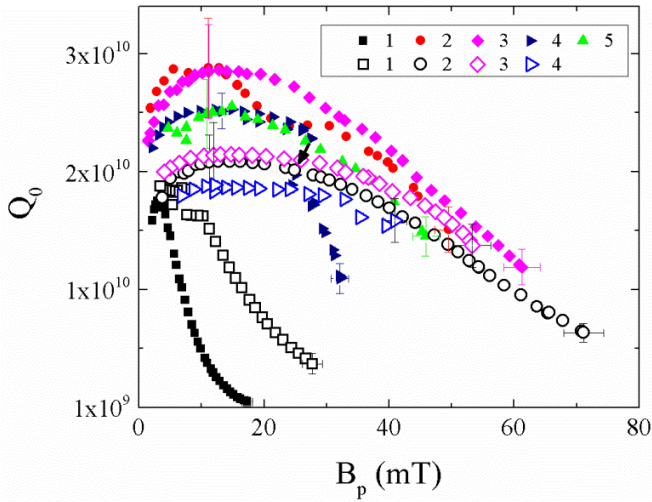


Figure 12. $Q_0(B_p)$ curves measured at 2.0 K for TC1N1 (solid symbols) and OCC1N2 (empty symbols) during each test, numbered in the legend. For both cavities, the RF performance was limited by a quench in all but the first test which was power limited. A “Q-switch” occurred at 30 mT during test 4 of TC1N1.

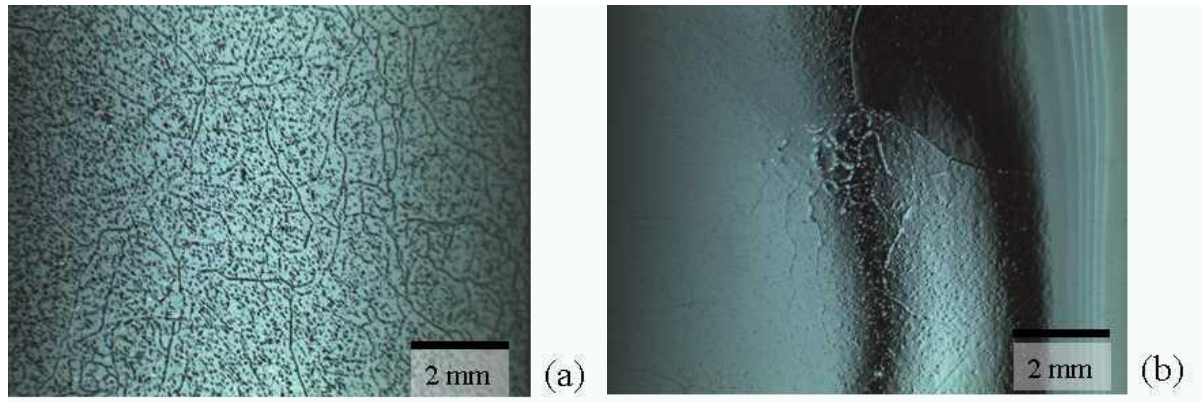


Figure 13. Images of the inner surface of TC1N1 after BCP in preparation for test 1 (a) and after CBP and multiple EP, after the last test (b).

Table 6. Fit parameters of $R_s(T)$ at 5 mT, $R_{BCS}(5 \text{ mT})$ at 4.3 K, and maximum peak surface magnetic field for each cavity test.

Test No.	Δ/kT_c	ℓ (nm)	R_{res} (n Ω)	R_{BCS} (n Ω)	$B_{p,max}$ (mT)
TC1N1					
1	1.80 ± 0.05	25 ± 300	13.0 ± 0.6	507 ± 110	20 ± 1
2	1.858 ± 0.001	26 ± 37	2.0 ± 0.1	490 ± 24	50 ± 2
3	1.840 ± 0.004	25 ± 55	2.3 ± 0.1	503 ± 16	61 ± 3
4	1.858 ± 0.001	26 ± 38	3.4 ± 0.1	490 ± 16	32 ± 1
5	1.850 ± 0.001	26 ± 46	4.8 ± 0.1	473 ± 20	46 ± 2
OCC1N2					
1	1.89 ± 0.03	125 ± 57	4.6 ± 0.3	650 ± 23	28 ± 2
2	1.86 ± 0.02	105 ± 28	3.0 ± 0.1	660 ± 21	72 ± 3
3	1.832 ± 0.006	41 ± 9	2.3 ± 0.1	645 ± 17	53 ± 3
4	1.86 ± 0.01	78 ± 15	4.9 ± 0.1	635 ± 16	41 ± 2

5. Discussion

Figure 14 shows a summary of the cavities' quench field as a function of the RRR of the ingot together with data on fine-grain cavities from Ref. [21]. EP often resulted in $\sim 20\%$ increase of the quench field, consistent with data on fine-grain cavities [21-23]. The data in figure 14 show that the quench field has no significant dependence on RRR for EP-treated cavities. A similar conclusion was achieved from test results of fine-grain and ingot Nb 9-cell 1.3 GHz cavities at DESY [24, 25]. The average quench field of more than thirty single- and multi-cell BCP-treated ingot Nb cavities of RRR ~ 260 was ~ 120 mT [26], similar to the results on fine-grain cavities shown in figure 14. The origin of the quench in the medium-purity ingot Nb cavities used for this study is not clear. Annealing does not change significantly the quench field or location, unlike EP.

The quench field of low purity ingot Nb cavities was severely limited by the occurrence of etch-pits. Preliminary studies on coupon samples from the same material showed pitting occurring already prior to any mechanical deformation, more severely after BCP than after EP. Depth profiles of impurities by time-

of-flight secondary ion mass spectrometry in such samples showed higher oxygen and hydrogen concentration within the pits, compared to non-pitted areas. A more systematic study of the occurrence of pitting in Nb coupons of different purity and etched by BCP and EP is underway. Previous studies on etch-pits in ingot Nb samples cut from a cavity showed a correlation between high density of pits and dislocations and the presence of hydrocarbon contamination within pits [27, 28].

As a comparison, the results of RF tests at 2 K of few cavities recently built at Jefferson Lab using low-purity fine-grain Nb also showed a significant scatter in the values of the quench field: 73 mT for a 1.5 GHz 5-cell cavity [29], 100 mT for a 650 MHz single-cell [30] and 26 mT for a 400 MHz single-cell cavity [31]. The nature of the defects limiting the performance of the cavities was not investigated.

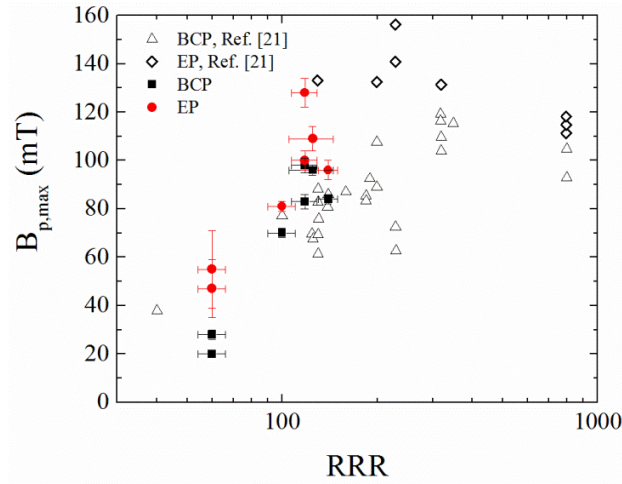


Figure 14. $B_{p,max}$ at 2.0 K as a function of RRR for medium and low purity ingot Nb cavities treated by BCP and EP (solid symbols). Empty symbols are data on fine-grain cavities at 1.5 K from Ref. [21].

The low-field BCS surface resistance at 4.3 K after BCP or EP is 500-650 n Ω at 1.3 GHz and 650-790 n Ω at 1.5 GHz, $\sim$ 20% lower than that of RRR>300 Nb [32] because of the lower mean free path ($\sim$ 100 nm compared to $\sim$ 600 nm) due to the higher concentration of interstitial impurities. An energy gap value of $\sim$ 1.45 meV was obtained after either BCP or EP, similar to that of high-purity Nb [32].

Figure 15 shows a summary of R_{res} as a function of RRR together with data on fine-grain cavities from Ref. [21]. The residual resistance values of $\sim$ 2-3 n Ω measured on low and medium ingot Nb cavities are comparable to the lowest values measured on fine-grain, RRR>300 cavities with similar treatments [21]. Additional data on multi-cell cavities at Jefferson Lab [33] and results at DESY, in which the quality factor of high-purity ingot Nb cavities was $\sim$ 20% higher than fine-grain ones for the same treatments [2] provide additional evidence of the possibility of achieving higher quality factors with ingot Nb material. One possibility to explain the reason for the lower residual losses could be the ability of the ingot material to expel residual magnetic field during cool-down below T_c more efficiently because of the lower density of grain boundaries [34] and weaker pinning force [35]. Except for the cavity H3H4, in which the half-cells were annealed at 1200 $^{\circ}$ C prior to welding, an average Q_0 -value of $\sim 2 \times 10^{10}$ at 2 K and 70 mT has been achieved with 1.3 GHz and 1.5 GHz medium-purity ingot Nb cavities, after BCP or EP only. This value is $\sim$ 70% higher than the average Q_0 -value measured at the same temperature and field on 7-cell, 1.5 GHz cavities for the upgrade of CEBAF, made of fine-grain, high-purity Nb with EP+LTB as final treatment.

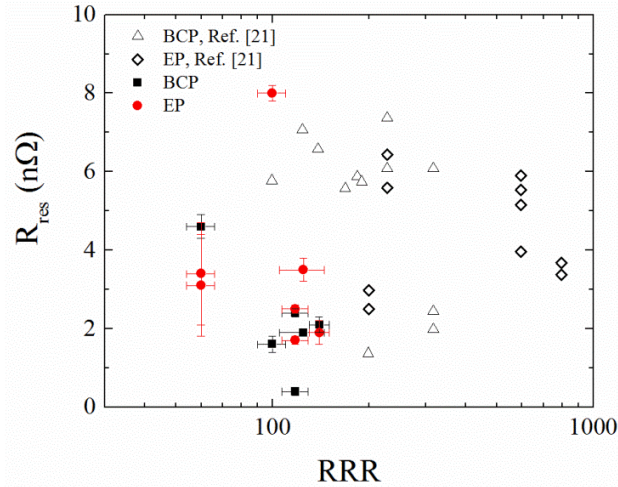


Figure 15. R_{res} as a function of RRR for medium and low purity ingot Nb cavities treated by BCP and EP (solid symbols). Empty symbols are data on fine-grain cavities from Ref. [21].

LTB reduces R_{BCS} , as a result of decreasing ℓ to ~ 30 nm and increasing Δ/kT_c by $\sim 4\%$ compared to BCP/EP treatments. LTB however increases R_{res} from an average of 2.6 n Ω to an average of 3.7 n Ω . The resulting average $Q_0(2.0$ K, 70 mT) after LTB is 2.2×10^{10} . The changes in material parameters resulting from LTB are consistent with those occurring in high-purity, fine-grain cavities [30]. HF rinse after LTB restores R_{res} to the values prior to LTB, without changing R_{BCS} , resulting in higher Q_0 -values at low field, consistent with results on high-purity, fine-grain cavities [36]. However, a stronger medium-field Q-slope after LTB makes the improvement of Q_0 -values at 70 mT only marginal and such Q-slope is not altered significantly by HF rinse. The average $Q_0(2.0$ K, 70 mT) after HF is 2.5×10^{10} . The high Q_0 -value in table 4 for cavity G1G2 after HF rinse was due to the Ti-doping done prior to the LTB and HF rinse and it is not included in the calculation of the average Q_0 after HF.

Doping medium-purity ingot Nb cavities either by nitrogen at 800 °C or by Ti above 1200 °C resulted in the even higher Q_0 -values but the quench field often dropped below 70 mT. A reduction of the quench field by doping has been observed in high-purity fine-grain and ingot Nb cavities as well and the possible causes are being studied by several laboratories [37-42]. The results so far suggest that the quench field is reduced more by increasing the partial pressure of the nitrogen gas. Because the RRR of the cavities used for this study is more than a factor of ~ 2 lower than that of high-purity ones, the cavities are already “partially” doped, so that significantly different process parameters might be required for doping medium-purity cavities to achieve a performance similar to that obtained for high-purity cavities after doping. In one instance, test 1 of cavity F3F4, the Q_0 increased with increasing RF field, up to 80 mT, as shown in figure 9 even though the cavity did not go through the doping process. It is possible that, after the surface preparation, the concentration of impurities already present in the material within the ~ 40 nm RF penetration depth, was within the proper range for the decrease of $R_s(B_p)$ to occur.

6. Conclusions

The results from the study presented in this article showed that:

- An average quench field of ~ 100 mT has been obtained in medium-purity (RRR ~ 120 -150) ingot Nb cavities at 2.0 K

- An average Q_0 -value of $\sim 2 \times 10^{10}$ at 2 K and 70 mT has been obtained in medium-purity, 1.3-1.5 GHz ingot Nb cavities because of the reduced R_{BCS} and R_{res} compared to high-purity, fine-grain cavities.
- Changes in low-field R_{BCS} and R_{res} by LTB and HF rinse are consistent with those obtained on high-purity, fine-grain Nb.
- The performance of low-purity ingot Nb cavities was limited by the occurrence of etch pits, particularly after BCP, and the cause for this is under investigation.

In summary, this study shows that medium-purity ingot Nb is a good material to build SRF cavities operating at medium gradients with higher efficiency and potentially lower cost than standard high-purity, fine-grain cavities. Two 5-cell cavities made from medium purity ingot Nb material are planned to be fabricated and tested at Jefferson Lab in 2016 as prototypes for a future refurbishment program of cryomodules installed in CEBAF to maintain the accelerator's beam energy of 12 GeV in the future years.

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